

SENG1050 Web Technologies

Lab 1

Part A*

1. List two major differences between LAN and WAN.

LAN: small in size; cheap and easy to install; geographically close.

WAN: large in size; expensive and requires extensive planning to install; geographically far apart.

2. Briefly describe 4 basic network topologies. Cite the advantage and disadvantage of each topology.

Bus: Allows multiple nodes to share a common communication medium; Crosstalk/collision may occur if multiple nodes want to transmit simultaneously (remedied through protocols such as CSMA/CA); Security concerns as all nodes can eavesdrop on the shared communication medium.

Ring: No collision as traffic flows in a single direction around the ring; Security concerns as traffic may be intercepted by non-destination nodes; Slow as the traffic need to travel around the entire ring for a node that might be adjacent to the source node; Network disruption if a node is not available.

Star: Network will not be disrupted if a non-central node is not available; Central node is a single point of failure.

Mesh: Fast and direct communication between all nodes; Expensive and impractical hardware setup.

Tree: Able to zone out network traffic; Similar disadvantages to bus networks.

3. What is the difference between a backbone link and a backbone network? How does this relate to the Internet Backbone?

Backbone link: Refers to a physical connection that carries most of the network traffic.

Backbone network: Collection of critical infrastructure/network devices that enable network traffic.

The Internet Backbone refers to the backbone network of the entire Internet. Undersea fiber-optic cables would be an example of a backbone link in this backbone network.

4. Explain the main difference between circuit switching and packet switching. Which switching technique does the Internet use?

Circuit switching: A direct connection needs to be established between two nodes before communication and each message is sent as a single payload.

Packet switching: Each message is split into multiple packets and relayed via multiple paths that are not directly established before communication.

The Internet uses datagram packet switching.

5. What is the difference between *forwarding* and *routing*?

Routing: The process of using a routing algorithm to determine the (best) route that can be taken by packets from the source to the destination. This usually results in a routing/forwarding table.

Forwarding: The process of moving packets from a router's input to its output by referring to a packet's header and a routing/forwarding table.

6. Explain what happens when router overrun occurs?

When router overrun occurs, extra packets are dropped and must be resent. Packet drops are detected when no acknowledgement (ACK) packets are received within the timeout period.

7. What is a protocol? Why does the Internet need to use protocols?

Protocol is a set of rules for communication. The Internet needs protocols to establish a common set of communication rules for otherwise incompatible networks.

8. What architecture does the Internet use?

TCP/IP architecture.

9. What does TCP stand for? What is the main service that the TCP provides?

Transmission Control Protocol. Provides reliable communication over packet-switched (unreliable) networks.

10. What does IP stand for? What is the main service that it provides?

Internet Protocol. Defines how packets are routed across the Internet.

11. What is an IP address? How many IP addresses can a router have?

An IP address is an identifier (32 bits for IPv4, 128 bits for IPv6) that uniquely and universally defines a node's connection to the Internet / a specific network. A router can have unlimited IP addresses based on the number of networks that it is connected to (LAN, WAN, isolated WiFi, VPN, virtual networks, etc). However, this may be limited by hardware capability and protocol version. Note that the minimum IP address is usually two, one facing LAN and one facing the Internet.

12. How many bits does an IP address have?

IPv4: 32 bits; IPv6: 128 bits.

13. Assuming that a data packet is going to be sent to a recipient with an IPv4 address of 201.22.32.10. How will this information be presented in the IP header of the data packet?

In 32 binary bits: 11001001 00010110 00100000 00001010. Note that an IPv4 address consists of four period-separated blocks. Each block is represented by a decimal number and we have to convert this to a 8-bit binary number (this can be done with a scientific calculator using decimal to binary conversion or by reading the remainder in reverse order when we divide the decimal number by 2).

14. Why does the Internet need DNS?

DNS provides a symbolic equivalent for numeric IP addresses. This allows numeric IP addresses to be represented as human memorable domain names.

15. What is the protocol for sending emails over the Internet?

Simple Mail Transfer Protocol (SMTP).

16. What application layer protocols will need to be involved from after you type "newcastle.edu.au" in your browser and press return, to when the browser displays the webpage?

DNS: A request is sent to one or more DNS servers to resolve "newcastle.edu.au" and obtain the corresponding IP address of the server.

HTTP: A GET request is sent to the IP address of the server. The server will return a response with the payload (a HTML file).

17. What is the Internet? Is the Internet actually the WWW? Why or why not?

The Internet is a network of networks (in terms of physical connection of network devices).

The Internet is not the WWW.

The WWW is one of the many services accessible over the Internet. The WWW is a distributed client/server service which allows any client with a browser to access services from servers.

Part B

1. Interpret the HTTP header of "<https://en.wikipedia.org/wiki/HTTP/3>" by using:

a. Command Prompt

i. Open the command prompt.

Windows: Click "Start" -> type "cmd" -> press "Enter"

Linux/Mac: Open any terminal app

ii. Enter the following on the command line:

```
curl -IL https://en.wikipedia.org/wiki/HTTP/3
```

b. Firefox, Chrome or Safari

i. Load the page "<https://en.wikipedia.org/wiki/HTTP/3>."

ii. Right click anywhere on the page, and select "**Inspect**" to open the Inspector

iii. Navigate to the **Network** tab, select the request that matches the URL you entered. Refresh the page if you cannot see it.

iv. Select "**Headers**".

Answer the following questions:

a) What is the version of HTTP used? (Hint: Right-click on a column name in the Inspector and enable the 'Protocol' column)

Command prompt: HTTP/1.1.

Browser: HTTP/2 (may differ according to browser).

b) What type of HTTP request has been used to fetch this page?

GET.

c) What status code can you see? What does that mean?

200 OK. Action was successfully received, understood, and accepted.

2. Repeat the steps and answer the questions in 1 (above) for '<https://en.wikipedia.org/wiki/HTTP/5>'. What are your observations?

Similar except for (3) where 404 Not Found is returned.

3. Find your local (private) IP address by using the following two methods:

Windows

1) Open the **Settings** app. Then go to **Network & Internet -> Advanced network settings**. From there scroll down and click on **Hardware and connection properties**.

2) Type `ipconfig/all` on the command prompt

Mac

1) Open **Network** in **System Preferences**

2) Type `ipconfig getifaddr en0` on Terminal if you're connected to WiFi, or `ipconfig getifaddr en1` if you're using Ethernet.

Linux

Type `ip address show` on the command line.

Answer the following questions:

a) What is your local (private) IP address?

This varies according to network class, subnetting and if private addresses are used.

b) Take note of the field 'DNS Servers'. What do you think these IP addresses represent?

IP addresses of immediate DNS servers that provide DNS lookups.

c) Look at the field 'Subnet Mask'. What do you think this field means?

Most significant bits which represent the network (subnet) portion of an IP address. This maps to x if we are using the CIDR notation of an IPv4 address: a.b.c.d/x.

4. Find your external (public) IP address by searching for "*what is my IP address*" on a search engine (verified on [google.com](https://www.google.com) and duckduckgo.com).

Note that this is different from the result of `ipconfig/all`. `ipconfig` shows internal (private) IP address. This means that routers usually have two IP addresses as a minimum, one internal and one external.

5. In the command prompt, type "ping google.com". What is the IP address? What do you think would be the result of entering this IP address into a browser's address bar? Try it, was the result what you expected?

Arrive at the same webpage. This demonstrates that DNS helps map human memorable domain names to IP addresses.